Huawei Access Node SOP: Resolving Optical Transport Failure on YEL-ACC-17 at Yelahanka Sub-station

Node: YEL-ACC-17 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_9961

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve a Major Optical Transport Failure on Huawei Access Node YEL-ACC-17 located at Yelahanka Sub-station. The failure is characterized by Alarm Code ALARM_9961, indicating a degradation in the SFP/XFP or fiber cuts, resulting in Rx/Tx power loss. The root cause of this failure is attributed to the Optical Transport Network (OTN) technology employed in the Huawei Access Node, which utilizes SFP/XFP transceivers to transmit and receive optical signals over fiber optic cables. The OTN technology is based on the ITU-T G.709 standard, which defines the protocol for transmitting and receiving optical signals over OTN networks. The failure can occur due to various reasons such as SFP/XFP degradation, fiber cuts, or Rx/Tx power loss, which can cause the OTN network to become unstable and generate the ALARM_9961 alarm. To resolve this failure, it is essential to diagnose the root cause and perform the necessary repairs or replacements to restore the OTN network to a stable state.

Pre-requisite Checks

- Check: Verify the power status of the Huawei Access Node YEL-ACC-17 at Yelahanka Sub-station using the display power command., Expected Outcome: The display power command should indicate that the node is powered on and operating normally.

- Check: Verify the redundant link status of the OTN network using the display ott link command., Expected Outcome: The display ott link command should indicate that the redundant link is active and operational.

- Check: Verify the SFP/XFP status of the Huawei Access Node YEL-ACC-17 at Yelahanka Sub-station using the display sfp command., Expected Outcome: The display sfp command should indicate that the SFP/XFP is operational and not degraded.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display power, Expected Output: Power status: ON,
Description: This command is used to verify the power status of the
Huawei Access Node YEL-ACC-17 at Yelahanka Sub-station. The expected
output indicates that the node is powered on and operating normally.

Command: display ott link, Expected Output: Link status: Active,
Description: This command is used to verify the redundant link status
of the OTN network. The expected output indicates that the redundant
link is active and operational.

Command: display sfp, Expected Output: SFP status: Operational,
Description: This command is used to verify the SFP/XFP status of the
Huawei Access Node YEL-ACC-17 at Yelahanka Sub-station. The expected
output indicates that the SFP/XFP is operational and not degraded.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP, Command: replace sfp slot 1/1-1/1, Description: This step involves replacing the degraded SFP/XFP with a new one. The replace sfp command is used to replace the SFP/XFP in slot 1/1-1/1. This command should be executed carefully to avoid any disruption to the OTN network.

Step 2: Step: Repair the fiber cut, Command: repair ott link 1/1-1/1, Description: This step involves repairing the fiber cut. The repair ott link command is used to repair the OTN link in 1/1-1/1. This command should be executed carefully to avoid any disruption to the OTN network.

Step 3: Step: Adjust the Rx/Tx power, Command: adjust ott power 1/1-1/1, Description: This step involves adjusting the Rx/Tx power. The adjust ott power command is used to adjust the OTN power in 1/1-1/1. This command should be executed carefully to avoid any disruption to the OTN network.

Post-Check Verification

```
Command: display ott link, Expected Output: Link status: Active,
Description: This command is used to verify the redundant link status
of the OTN network. The expected output indicates that the redundant
link is active and operational.

Command: display sfp, Expected Output: SFP status: Operational,
Description: This command is used to verify the SFP/XFP status of the
Huawei Access Node YEL-ACC-17 at Yelahanka Sub-station. The expected
output indicates that the SFP/XFP is operational and not degraded.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: restore sfp slot 1/1-1/1, Description: This step involves restoring the original SFP/XFP. The restore sfp command is used to restore the SFP/XFP in slot 1/1-1/1. This command should be executed carefully to avoid any disruption to the OTN network.

- Step: Restore the original OTT link, Command: restore ott link 1/1-1/1, Description: This step involves restoring the original OTT link. The restore ott link command is used to restore the OTT link in 1/1-1/1. This command should be executed carefully to avoid any disruption to the OTN network.